

Teaching Note

Design Thinking

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1. Platform Overview

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes the Introduction section, observe, Empathize, Define, ideate, Prototype, Execute, Decision Checklist, and Reports. Some sections must be completed first as they have an impact on other areas.

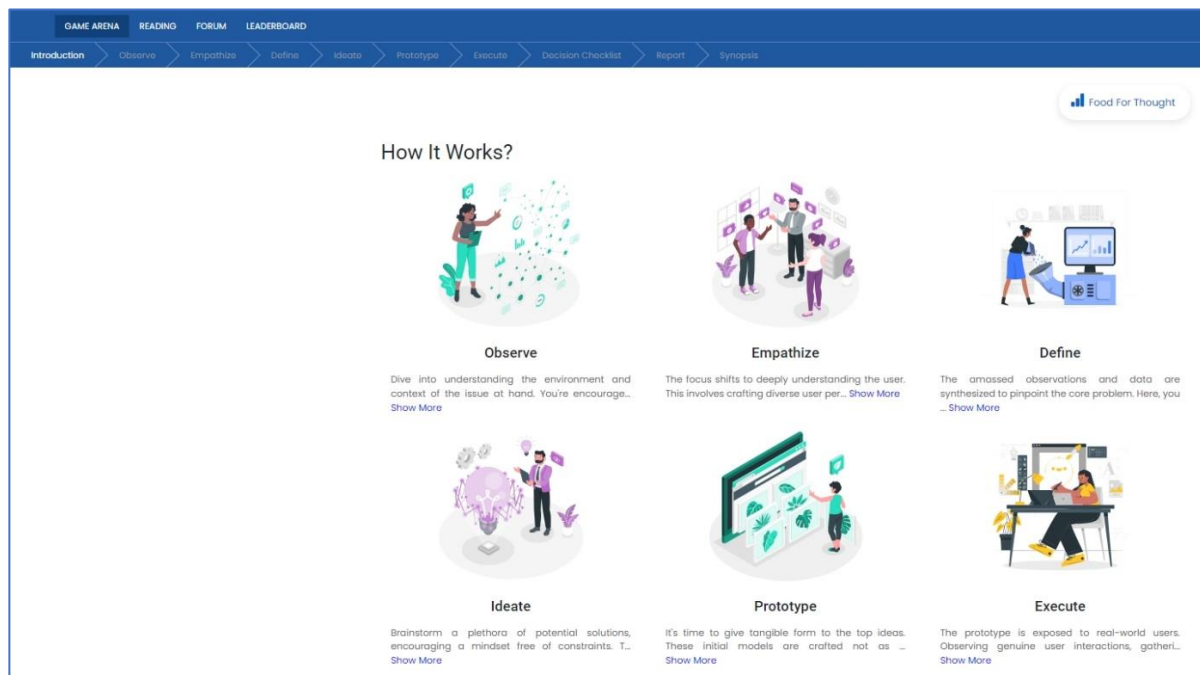


Figure 1 The Game Arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the case scenario, industry overview and the diverse array of decisions required.

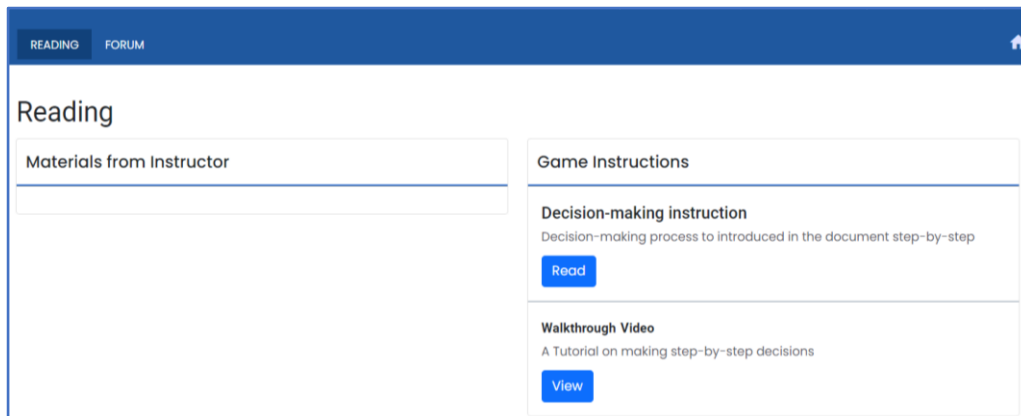


Figure 2 The Readings Section

1.3 Forum

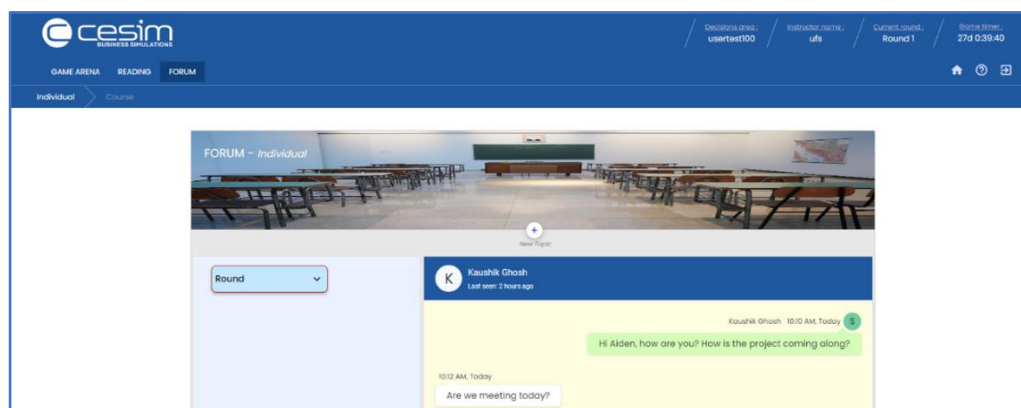


Figure 3 The Forum Section

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

Tech-Time Innovations, a decade-old electronics company founded by engineers from premier institutions, has established a strong presence with its fitness trackers and IoT devices. The company achieved a 20% market share for fitness trackers in urban India within three years and a 30% adoption rate for IoT devices in tier-1 cities, with Bengaluru and Mumbai leading the way. Customer demand highlights the need for an all-in-one wearable focused on health, fitness, and productivity.

Building on this success, Tech-Time is now expanding into the rapidly growing smartwatch market. Participants are tasked with identifying the opportunities Tech-Time Innovations can leverage and the challenges it may encounter in developing a user-centric smartwatch that resonates with Indian consumers. The end goal is to increase market share by 2% and achieve a profit margin of 8%.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

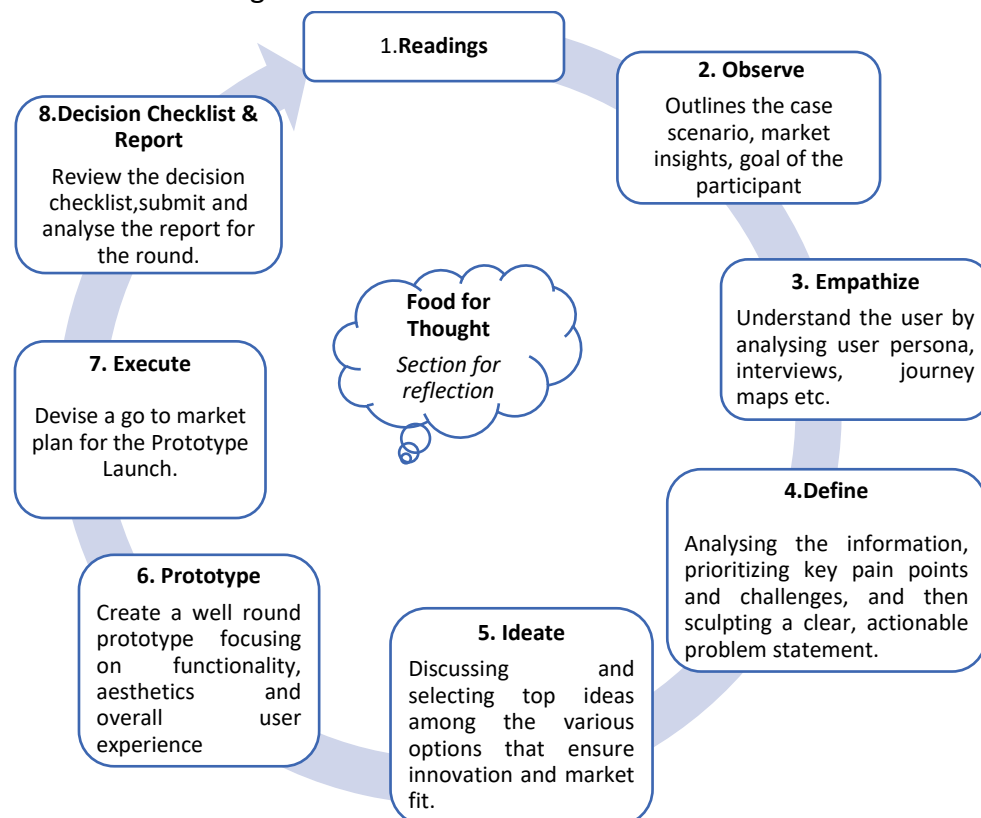


Figure 4 The Sequence of Decision-Making Process

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

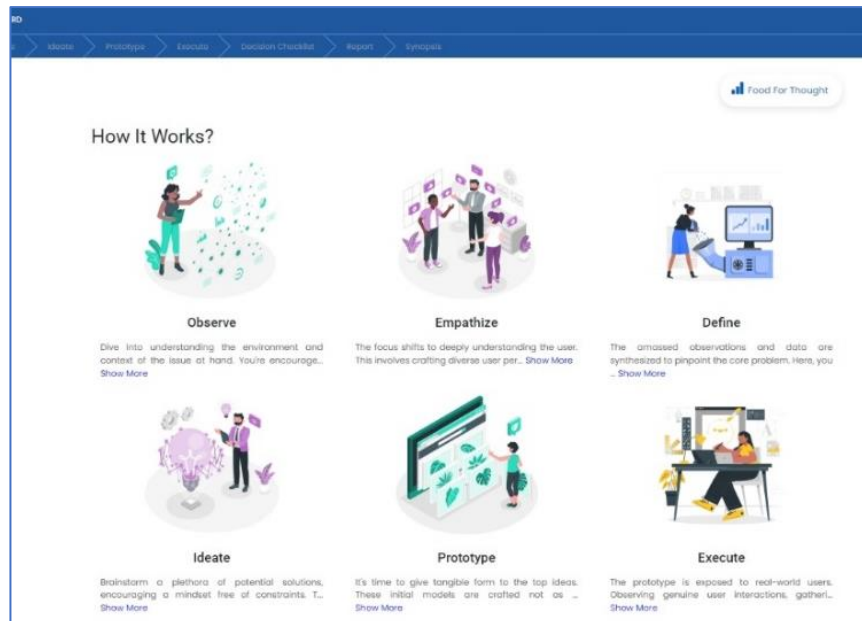


Figure 5 The Introduction Tab

2.2 Food for Thought

The "Food for Thought" section contains questions related to key decision areas, providing valuable insights to guide decision-making. Participants are encouraged to review it before addressing the decision areas, as it offers essential clarity on the decision-making process. Completion of this section is mandatory for Round one.

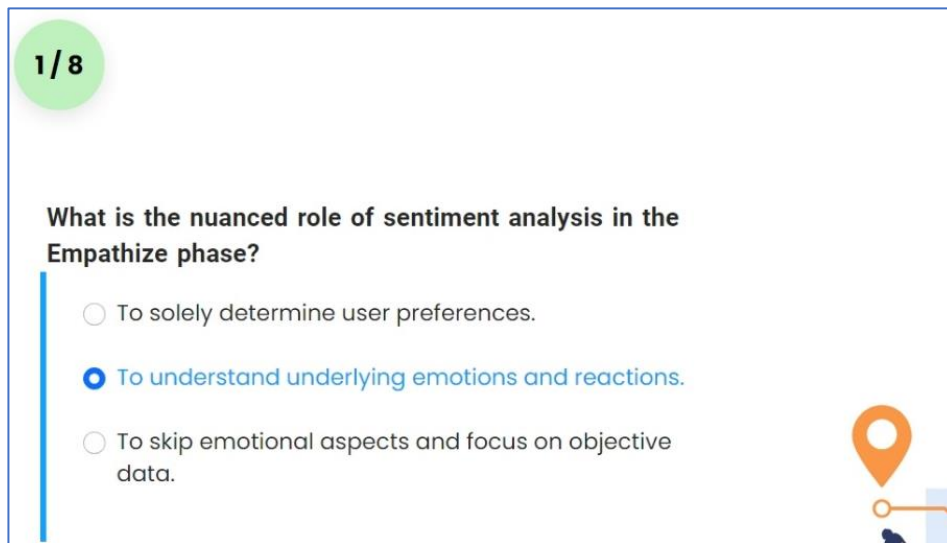


Figure 6 The Food for Thought Section

2.3 Observe

The **Observe Phase** in Design Thinking is focused on understanding the target users and market context by gathering insights, identifying trends, and observing behaviours and needs. In the Observe Phase, teams collect data from users, market research, and competitor analysis to identify opportunities, anticipate challenges, and ensure the product aligns with real user demands and market dynamics.

Summary of the Observe Phase in the Simulation

TechTime Innovations, a company with a decade of experience in the electronic market, is expanding into the smartwatch sector. The company's previous success includes a 20% market share in India's urban cities with fitness trackers and a 30% adoption rate for IoT devices in tier-1 cities, particularly in Bengaluru and Mumbai.

Customer feedback emphasizes a desire for an all-in-one wearable device focusing on health, fitness, and productivity. Market insights reveal that India's wearable market, especially the smartwatch sector, is growing rapidly, with a 30% year-on-year increase. The primary consumers are aged 18-35, who account for 60% of smartwatch sales. There is also a rising interest in health and fitness, driven by urban lifestyles and awareness campaigns.

The market features competition from international players like Apple and Samsung, as well as domestic brands such as Noise and Boat. Consumers are seeking a balance of price, features, and aesthetics. Research from a survey of 1,000 potential consumers shows that 70% prefer smartwatches focused on health and fitness. Additionally, 50% are interested in localized features, including Indian payment gateway integration and regional language support, while 40% prioritize longer battery life. The goal is to identify opportunities and challenges in creating a user-centric smartwatch for the Indian market.

2.4 Empathize

The **Empathise Phase** is the initial stage in the **Design Thinking** process, centered on gaining a deep understanding of users' requirements, emotions, and obstacles. Its goal is to gather valuable insights and foster empathy, ensuring that the design outcome aligns with users' true needs. This phase utilizes **User Personas**, **User Interviews**, **User Journey Maps**, and **Sentiment Analysis** to uncover important details about users' needs, pain points, and motivations. These methods enable designers to develop a comprehensive understanding of users, ensuring that the design process remains focused on the user and that the final solution is precisely tailored to address their real challenges and aspirations.



Figure 7 Empathize tab in the Game Arena

Key Sections of Empathise Phase

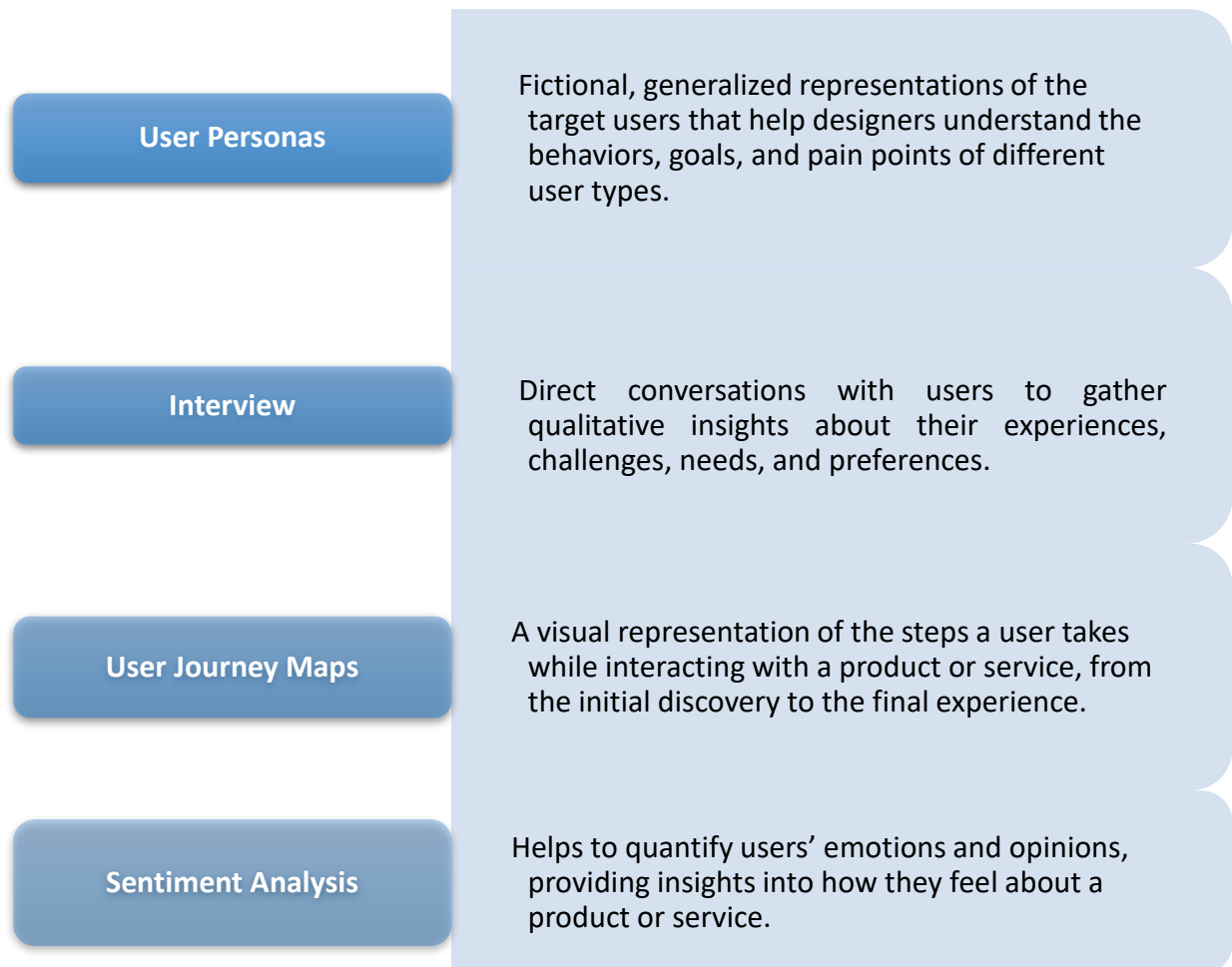


Figure 8 Key Sections in the Empathise Tab

2.5 Define

In the **Define** stage, participants review the **SWOT analysis** and **Gap analysis** to shape a clear and actionable product strategy. These tools bring clarity to the problem space by identifying strengths, weaknesses, gaps, and limitations, allowing the team to develop a focused problem statement, prioritize essential features, and set realistic goals. This approach ensures the design process addresses key challenges within feasible limits, aligning with user needs and practical constraints.

Based on this analysis, participants make critical decisions regarding battery life, affordability, and localization features, ensuring the product meets user needs and aligns with market demands. This strategic focus enhances the product's relevance and success potential by aligning design choices closely with user expectations and industry standards.

SWOT Analysis ?

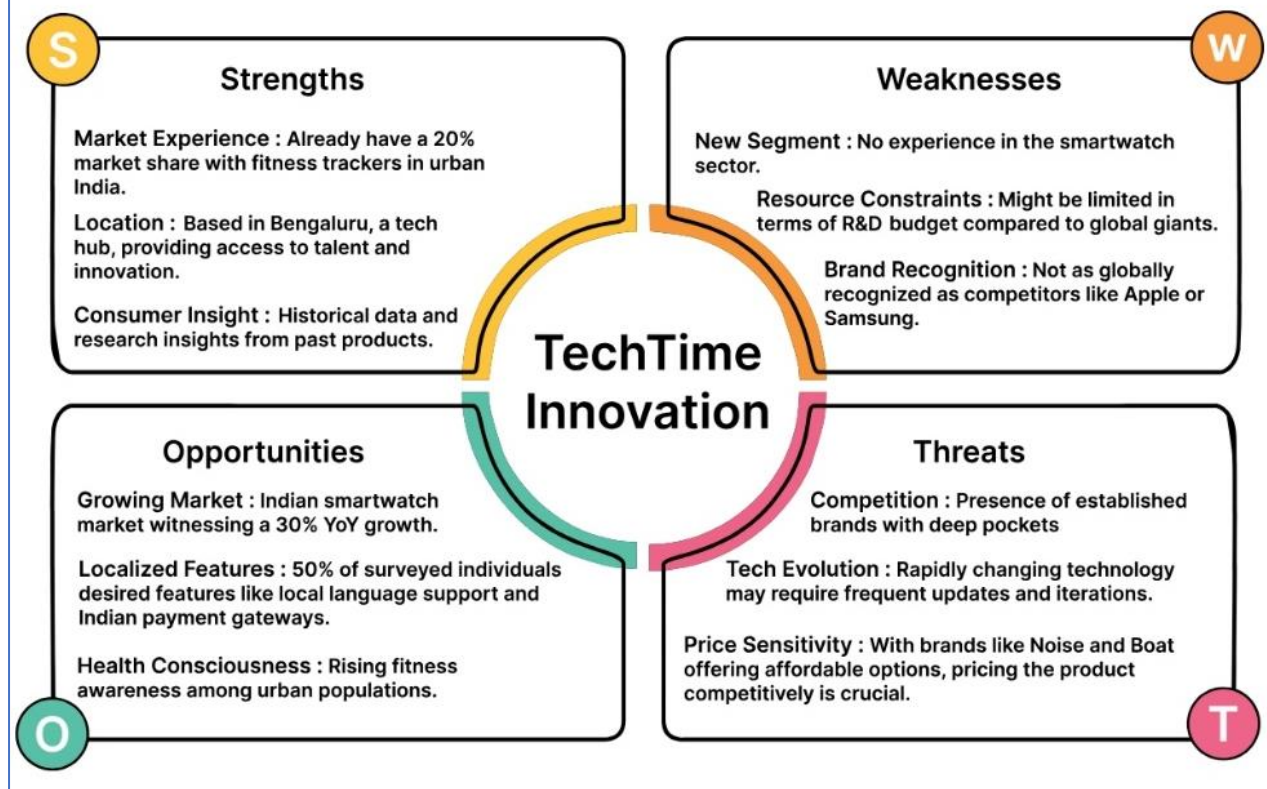


Figure 9 The Swot Analysis Provided in the Define Tab



Figure 10 The define tab in the Game Arena

2.6 Ideate

In this tab, participants are tasked with selecting the top three product ideas from the ten available options. While making their selections, they need to consider the features offered, the target segment, the closest competitors, and the rationale for why each product meets consumer requirements. This thoughtful approach ensures the chosen ideas align with market needs and competitive dynamics.

Ideate > Prototype > Execute > Decision Checklist > Report > Synopsis

Food For Thought

Ideate

TechTime Vital+

Features : Advanced health metrics, Ayurvedic wellness insights, ECG, regional health advisories.

Target Segment : Health-conscious adults, especially from urban areas like Bengaluru.

Closest Competitor : Apple Watch Series 6 (Market Share: 30%, Revenue: INR 15,000 Crore).

Context : Rising health consciousness among urban Indians.

TechTime Linguist

Features : Support for multiple Indian languages, instant translation, cultural event reminders, advisories.

Target Segment : Multilingual professionals and seniors who prefer regional languages.

Closest Competitor : Garmin Venu (Market Share: 10%, Revenue: INR 3,500 Crore).

Context : 50% of users desired localized features.

TechTime Eco

Features : Solar-powered, carbon footprint tracker, sustainability tips tailored for Indian users.

Target Segment : Environmentalists and the young urban population.

Closest Competitor : Garmin Fenix 6X Pro Solar (Market Share: 8%, Revenue: INR 2,800 Crore).

Context : Sustainability as a rising global and national trend.

Figure 11 The Ideate Tab in the Game Arena

Comparison of Product ideas available under the Ideate Phase :

	Product	Features
1	TechTime Vital+	High demand for health features, combines advanced health metrics with Ayurvedic insights—addresses a clear gap in the market.
2	TechTime Linguist	High appeal due to support for Indian languages, translations, and cultural reminders, aligning with localization demands.
3	TechTime DesiFit	Unique focus on Indian fitness trends and diet; culturally specific features give it an edge in the fitness segment.
4	TechTime Eco	Sustainability is a rising trend globally and in India; this eco-friendly concept would appeal to environmentally conscious users.
5	TechTime PayBharat	UPI and local payment integration is highly relevant in urban India; it could drive adoption among shoppers and professionals.
6	TechTime Yatri	India-specific travel and festival guidance has appeal, though it is niche; strong potential among tourists and urban travelers.
7	TechTime KidGuard	Strong appeal for parents and families due to safety and educational value but limited in age-specific scope.
8	TechTime Dharma	Unique spiritual features could attract niche users interested in wellness, though market size is relatively limited.
9	TechTime Trendsetter	Fashion and design elements are appealing to youth but may lack distinctive functional advantages in a competitive market.
10	TechTime Academia	Good focus on academic tools, but limited audience compared to other segments; study aids may be less compelling in a smartwatch.

2.7 Prototype

The **Prototype Phase** in design thinking is where initial concepts are transformed into tangible models or representations, allowing teams to explore how well these ideas work in practice. This phase is crucial because it translates abstract ideas from the ideation phase into concrete experiences that can be tested, refined, and improved based on user feedback.

In this phase, participants focus on developing prototypes of their chosen ideas. They make important decisions regarding the features to include in the prototypes, considering various aspects such as **aesthetics, battery life, language support, water resistance, health and technology integration etc.** Each feature choice is aimed at aligning the prototype with the preferences and expectations of the target users while ensuring it stands out in the market.

While selecting features, participants must also account for the **fluctuating costs associated with prototype development** and the **affordability** of the product for the intended target group. Maintaining the **value proposition**—the unique benefit the product promises to deliver—without compromising on quality is key. Cost-effective decision-making in the prototype phase helps avoid over-designing or including features that may raise costs without adding significant value to users.

Ultimately, the Prototype Phase is an iterative process where ideas are tested, refined, and validated to ensure they resonate with users' needs and are feasible to produce. It enables teams to evaluate the feasibility of their designs and prepare for the next stage: user testing and further refinement. This phase is essential for creating user-centered solutions that are both innovative and practical.

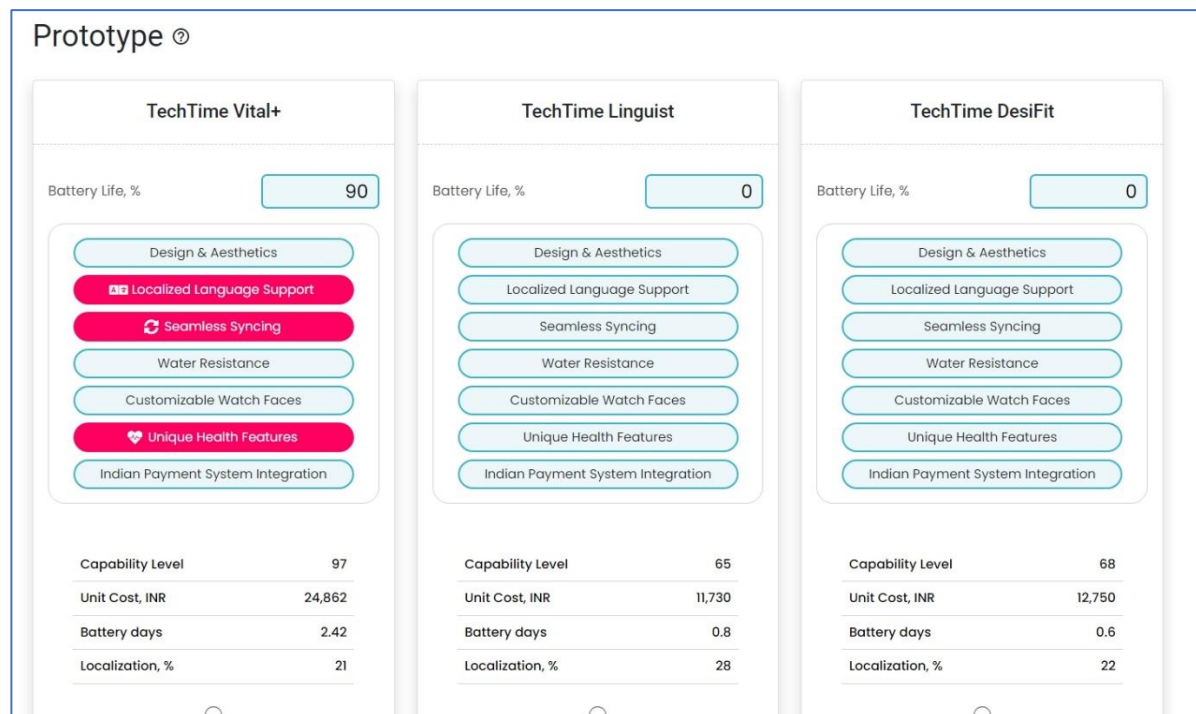


Figure 12 The Prototype Tab in the Game Arena

2.8 Execute

The Execute Phase in Design Thinking is the stage where the ideas and concepts that have been developed through earlier phases (Empathize, Define, Ideate, and Prototype) are brought to life and implemented in the real world. This phase involves turning the validated prototypes into actual products and launching them into the market. The goal is to translate the insights, prototypes, and decisions from the previous stages into a functional product that meets the needs of the users and delivers on the brand's promises.

Key decisions about the marketing mix (price, advertising, warranty, and channel strategy) play a crucial role in how the product will perform in the market. Selecting the appropriate sales channels is particularly important as it determines how effectively the product will reach and engage the target audience.

Key Aspects of the Execute Phase

In the Execute Phase, participants are tasked with making critical decisions related to the **marketing mix** for the product, which includes:

1. **Decision on Price:** In this section, participants focus on establishing a price that resonates with their target audience. Setting an effective price requires balancing the product's value proposition with the purchasing power of the target market. The price should not only be competitive within the market but also reflect the unique benefits the product offers. Key considerations include covering production costs, achieving a desired profit margin, and ensuring the price aligns with customer expectations and market conditions. This thoughtful approach helps position the product successfully while driving profitability.
2. **Allocation of Advertising Budget:** In this section, participants are tasked with allocating the advertising budget. They can refer to the graph adjacent to the decision area, which displays promotional spending trends over recent years among global giants, mid-sized international companies, and local players. By analysing these patterns, participants can identify competitive spending benchmarks and allocate their budget effectively to maximize visibility and reach within their target market.

3. **Warranty Period:** In this section, participants are tasked with determining the optimal warranty period for the product. The warranty duration should strike a balance between meeting consumer expectations and the company's capacity to support the product effectively. A robust warranty can enhance the product's appeal, serving as a valuable selling point that builds consumer trust. The adjacent graph illustrates the relationship between the warranty period and the product's attractiveness score, providing insights to guide this decision.

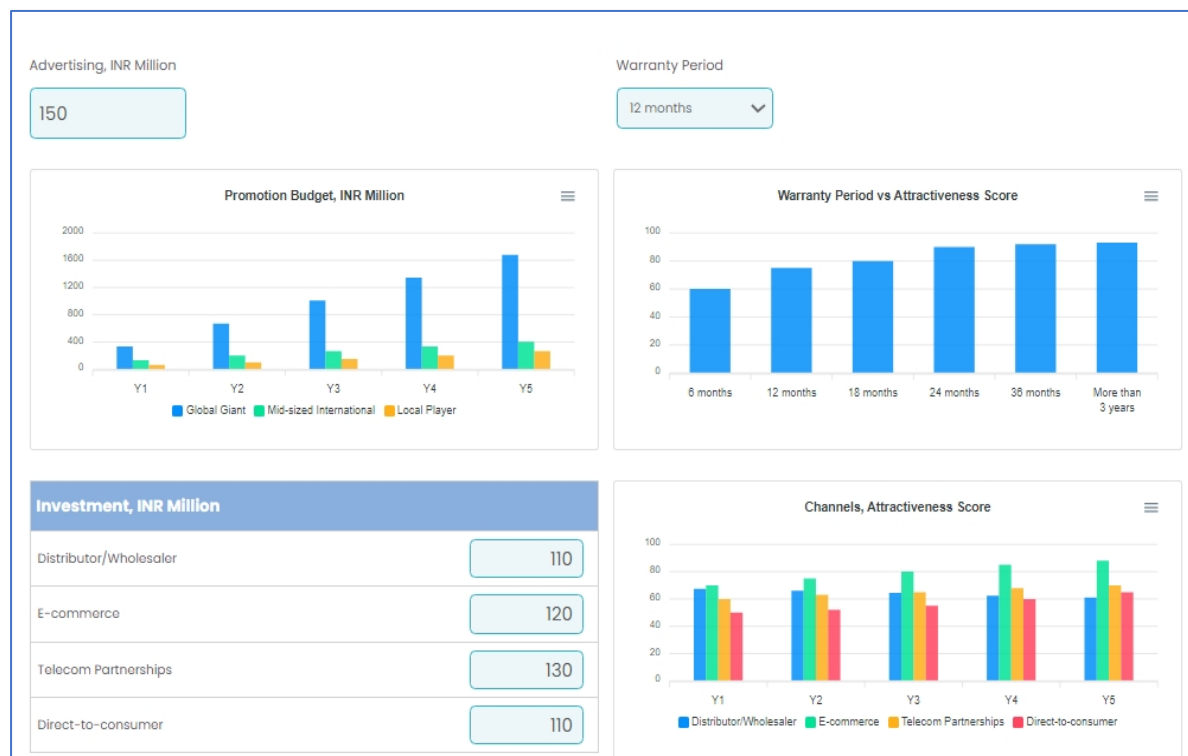


Figure 13 The Execute Tab in the Game Arena

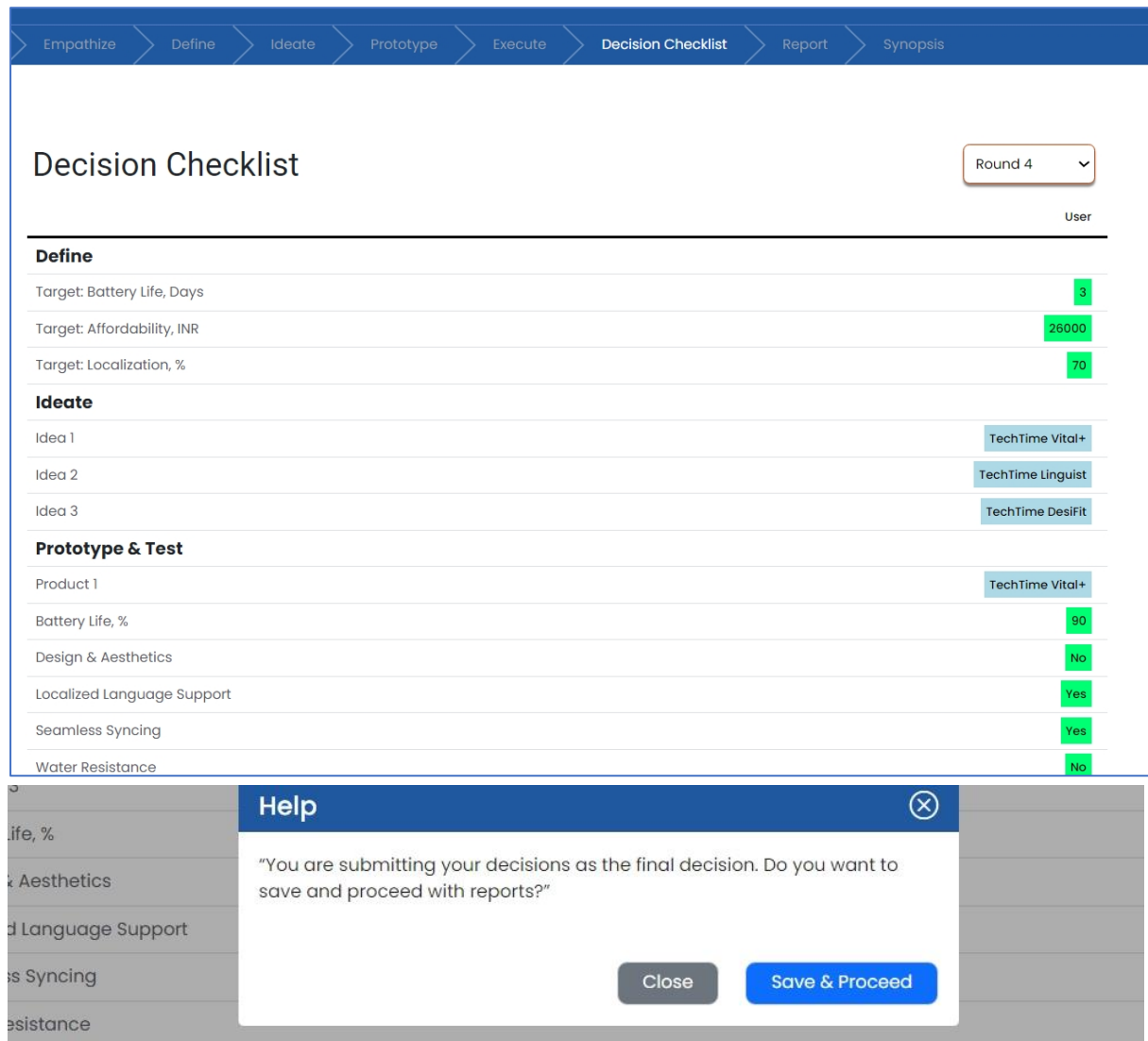
4. **Investment - Channel Budget:** In this section, participants are tasked with determining which sales channels to prioritize for product distribution and allocating the budget accordingly. The "Channel Attractiveness Scores" graph provided in the decision area offers insights into which channels have been most appealing to the audience over the years. Prioritizing the right channels is essential for making the product accessible to the target audience and delivering an optimal customer experience.

There are four primary sales channels to consider when executing the product launch:

1. **Distributor/Wholesaler:** This channel involves selling products in bulk to distributors or wholesalers who then sell them to various retailers. It enhances **market reach** and **efficiency**, particularly for products intended to be widely available across different regions.
2. **E-commerce:** Selling through **online platforms**, or the company's own website offers access to a broad **customer base**. This channel is cost-effective as it bypasses the need for physical retail spaces, allowing for lower **operational costs**. Additionally, e-commerce platforms offer valuable data on **consumer behavior**, helping the company refine its marketing and product strategies based on customer preferences, purchase history, and feedback.
3. **Telecom Partnerships:** Collaborating with **telecom companies** offers a strategic avenue to bundle products with mobile plans, which can significantly boost visibility and accessibility. Telecom companies have an established customer base, and their partnership can create opportunities for **cross-promotions** or special offers, making the product available to a wide audience.
4. **Direct-to-Consumer (D2C):** Selling **directly to consumers** through a brand's own platform (online or offline) gives the company full control over the **brand experience**. This method improves **profit margins** because there is no intermediary involved, and it enables the company to maintain a strong relationship with customers. It also allows for personalized marketing, greater customization of the shopping experience, and direct collection of customer feedback.

2.9 Decision Checklist

The Decision Checklist helps participants review and finalize their decisions. Once decisions are confirmed, they can submit them using the "Submit" option and choose [save and proceed](#). It's important to note that once decisions are submitted and saved, they cannot be altered for that round. This ensures that decisions are carefully considered before finalization, promoting clarity and accountability in the decision-making process.



Decision Checklist		Round 4	User
Define			
Target: Battery Life, Days		3	
Target: Affordability, INR		26000	
Target: Localization, %		70	
Ideate			
Idea 1		TechTime Vital+	
Idea 2		TechTime Linguist	
Idea 3		TechTime DesiFit	
Prototype & Test			
Product 1		TechTime Vital+	
Battery Life, %		90	
Design & Aesthetics		No	
Localized Language Support		Yes	
Seamless Syncing		Yes	
Water Resistance		No	

Help

"You are submitting your decisions as the final decision. Do you want to save and proceed with reports?"

Close Save & Proceed

Figure 14 The Decision Checklist in the Game Aren

2.10 Report

- The report section provides the details on the participants' critical thinking abilities. It demonstrates how their thought process influenced the outcomes, providing insights into the impact of their decision-making skills.

This section consists of feedback on the performance of the participant based on four metrics.

- Rigour: Consistency throughout the decision-making and recommendation.
- Structuring: Logical approach of gathering and analysing information
- Synthesis: Approach towards interpreting information and making informed decisions
- Business Judgement: Ability to evaluate facts, and consider risks and possible implications when making decision.

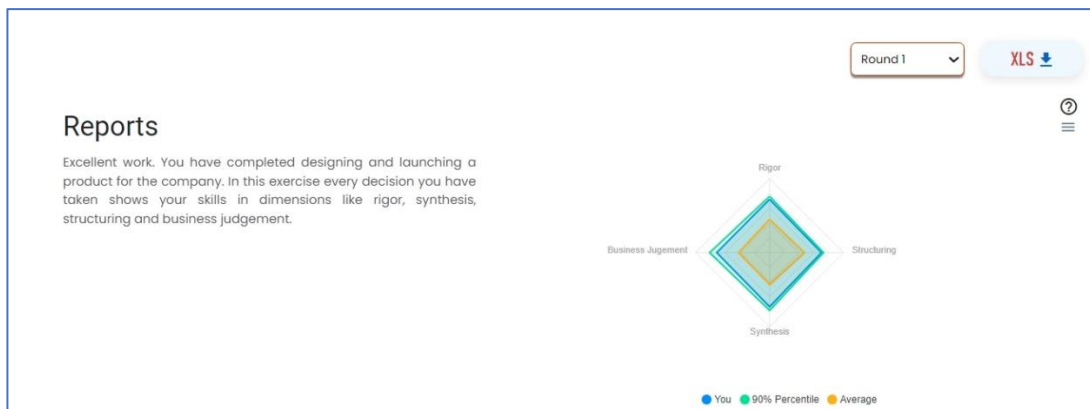


Figure 15 The Report in the Game Arena

The report includes the income statement, product details, and key performance indicators (KPIs). The income statement provides a detailed breakdown of revenue generated from product sales, as well as various costs associated with production, administration, and marketing efforts, culminating in the final operating profit or loss. An adjacent graph offers a visual summary of these financials.

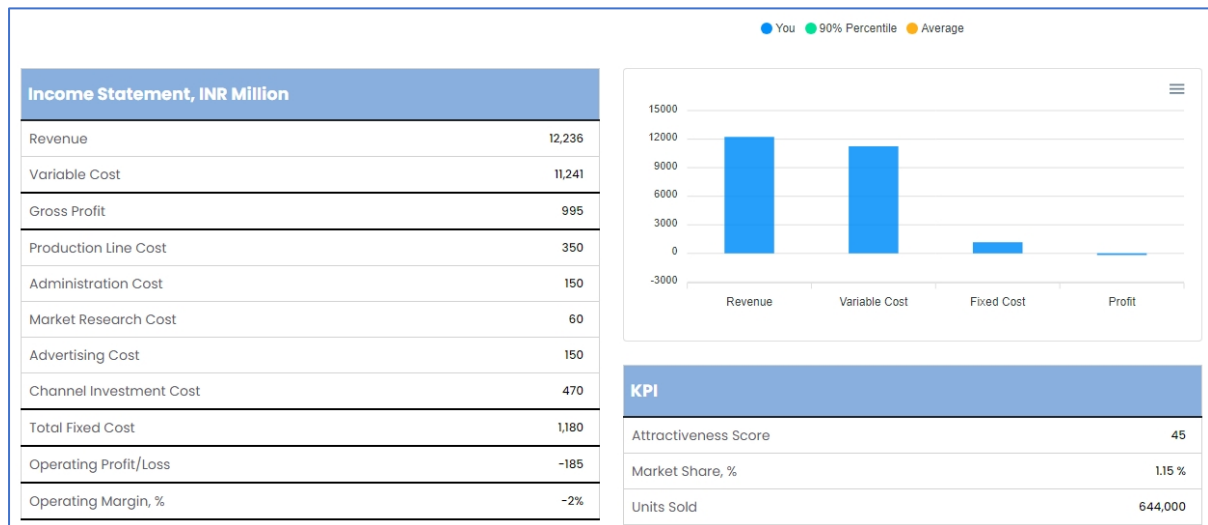


Figure 16 The Income Statement- Report in the Game Arena

The product details section outlines key decisions made about the product, including added features, target segment, closest competitors, and alignment with market needs. Additionally, it offers strategic tips for product and market expansion.

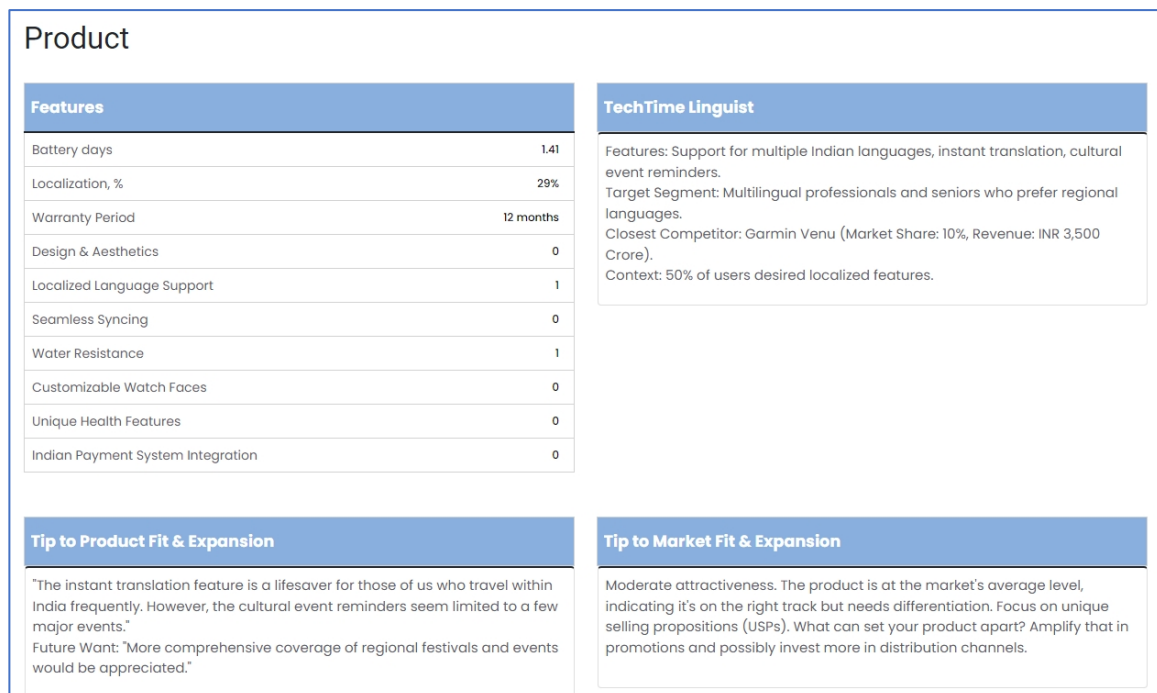


Figure 17 The Product Details - Report in the Game Arena

Participants are encouraged to identify what differentiates their product, highlight these attributes in promotional efforts, and consider increasing investment in distribution channels to strengthen market presence. For upcoming rounds, participants should aim to improve key performance indicators (KPIs), enhance operating margins, and expand market share. The end goal is to increase market share by 2% and achieve a profit margin of 8%.

2.11 Synopsis

The synopsis section provides a condensed overview of the key decision areas and outcomes. It offers participants a quick reference point to grasp the essence of their decisions and their thinking ability, aiding in holistic understanding and future planning.

Synopsis		Round 1	XLS
Input		Income Statement, INR Million	
Parameters	Input	Parameter	Output
Target: Battery Life, Days	3	Revenue	12,236
Target: Affordability, INR	16000	Variable Cost	11,241
Target: Localization, %	70%	Gross Profit	995
Idea 1	TechTime Linguist	Production Line Cost	350
Idea 2	TechTime Dharma	Administration Cost	150
Idea 3	TechTime PayBharat	Market Research Cost	60
Product 1	TechTime Linguist	Advertising Cost	150
Battery Life	60%	Channel Investment Cost	470
Design & Aesthetics	No	Total Fixed Cost	1,180
Localized Language Support	Yes	Operating Profit/Loss	-185
Seamless Syncing	No	KPI	
Water Resistance	Yes	Parameter	Output
Customizable Watch Faces	No	Attractiveness Score	45
Unique Health Features	No	Market Share, %	1%
Indian Payment System Integration	No	Units Sold	64
Product 2	TechTime Dharma	Operating Margin, %	-2%
Battery Life	70%		
		Round 2	

Figure 18 The Synopsis in the Game Arena

To proceed to the next iteration, Navigate to the end of the synopsis page to locate the link for the next round and Click on the 'Round' to advance to the next round.

2.12 One of the Ideal Solution

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution is just one example for faculty reference and not intended for direct communication to students.

Decision Area	Input
Target: Battery Life, Days	2
Target: Affordability, INR	20,000
Target: Localization, %	40%
Ideate	
Idea 1	TechTime Linguist
Idea 2	TechTime PayBharat
Idea 3	TechTime Academia
Prototype & Test	
Product 1	TechTime Linguist
Battery Life	80%
Design & Aesthetics	No
Localized Language Support	No
Seamless Syncing	No
Water Resistance	Yes
Customizable Watch Faces	Yes
Unique Health Features	Yes
Indian Payment System Integration	No
Product 2	TechTime PayBharat
Battery Life	70%
Design & Aesthetics	Yes
Localized Language Support	Yes
Seamless Syncing	Yes
Water Resistance	Yes
Customizable Watch Faces	Yes
Unique Health Features	No
Indian Payment System Integration	No
Product 3	TechTime Academia

Battery Life	70%
Design & Aesthetics	Yes
Localized Language Support	Yes
Seamless Syncing	No
Water Resistance	Yes
Customizable Watch Faces	No
Unique Health Features	No
Indian Payment System Integration	No
Product Launch	TechTime Linguist
Execute	
Price, INR	23,000
Advertising, INR Million	300
Warranty Period	6 months
Distributor/Wholesaler, INR Million	100
E-commerce, INR Million	300
Telecom Partnerships, INR Million	100
Direct-to-consumer, INR Million	100

Logic for Decision making

a) Product Overview and Target Segment

TechTime Linguist is a wearable device with features specifically designed for the Indian market, including multi-language support, instant translation, and culturally relevant reminders. It targets multilingual professionals and seniors who prefer interacting in regional languages. With 50% of potential users indicating a preference for localized features, TechTime Linguist addresses an important gap not fully covered by competitors like Garmin Venu, which holds a significant market share of 10% and generates substantial revenue in India. This product differentiation based on language support and cultural relevance is central to positioning TechTime Linguist as a unique option.

b) Product Development Choices in Prototype Phase

Battery Life (80% Capacity):

A balance was struck between functionality and power efficiency, with an 80% battery capacity enabling reasonable usage without frequent recharging, which is critical for convenience. Extensive battery life aligns well with the preferences of multilingual professionals and seniors, who may not be inclined to charge devices frequently. By offering battery life that supports prolonged usage, TechTime Linguist enhances user convenience without compromising on core features. This feature also competes favorably with alternatives, ensuring the product remains attractive to users who prioritize practicality.

Core Features (Water Resistance, Customizable Watch Faces, Health Monitoring):

These features were chosen to appeal broadly to the target demographic. Water resistance enhances durability, customizable watch faces add a personal touch, and health-monitoring features are increasingly in demand across age groups, particularly among seniors who may use the watch for basic health tracking. These features add functional value to the product, expanding its appeal and increasing its competitiveness. Offering health tracking at a competitive price point also provides an edge over premium competitors like Garmin Venu, allowing TechTime to differentiate itself as both a cost-effective and feature-rich alternative.

Unit Cost and Affordability

The decision to maintain a lower unit cost was aimed at ensuring affordability for the target market segment, which may be price-sensitive. By optimizing costs, the company can offer an accessible price that resonates with multilingual professionals and seniors. The lower unit cost allows for competitive pricing while still maintaining profitable margins. This pricing strategy aligns with the brand's goal of reaching a wide audience, increasing potential market share by positioning TechTime Linguist as a high-value yet affordable product.

c) Marketing Strategy and Budget Allocation

Advertising Budget (Mid-Sized Company Range)

Setting an advertising budget in line with mid-sized companies allows for impactful promotions without overextending financial resources. This budget range supports targeted marketing efforts while allowing room to allocate funds to other essential areas like product development and distribution. By optimizing the advertising spend, TechTime can maintain financial health while reaching the desired market segments through carefully selected channels. This budget level is likely to be sufficient for digital and traditional channels, ensuring comprehensive outreach and brand visibility.

Warranty Period (6 Months)

A six-month warranty period was chosen to balance product assurance with financial feasibility. While a longer warranty might appeal to some customers, the six-month period aligns with industry norms for similarly priced electronics. This warranty period gives customers confidence in the product's quality while managing the company's liability for potential repairs or replacements. It's an acceptable compromise for the target audience and matches expectations in this product category.

Channel Investment (Higher Allocation to E-Commerce)

A higher investment in e-commerce channels was chosen to maximize reach and accessibility for the target demographic, who are likely to shop online for electronics. E-commerce is cost-effective and can be easily scaled up as needed. This decision optimizes market penetration, particularly for tech-savvy multilingual professionals and seniors who may prefer the convenience of online shopping. With reduced overall channel spending, TechTime can improve margins while maintaining a solid distribution network. Allocating an equal budget to other channels ensures broad market coverage without over-investing, which allows for efficient resource allocation and cost control.

3. Implementation – Methodologies

From our experience, we have seen Professors using two different teaching plan options. Option A is for the one who is pressed on time while option B offers a chance to participants to use the module multiple times and learn from their past decisions.

Option A

To begin, the instructor should introduce the module with the walkthrough video, explaining to the students all the different decision-making points involved in the module and how to connect them with one another (with the use of teaching notes provided by Cesim). The instructor should then provide the opportunity for the students to play the module by themselves. The students can play the module within a team or individually. Once the module is over, the instructor can quickly review the decision-making made by the students along with the results they achieved. The instructor can compare the results of different students across the module and then debrief them while driving a discussion in the class.

1. Introduction (5 minutes)
2. Run Simulation (30 to 35 minutes)
3. Debrief (10 minutes)
4. Conclusion (8 to 10 minutes)

Option B

To begin, the instructor should introduce the module with the walkthrough video, explaining to the students all the different decision-making points involved in the module and how to connect them with one another (with the use of teaching notes provided by Cesim). The instructor should then provide the opportunity for the students to play the module by themselves. The students can play the module within a team or individually. The instructor should debrief them in short. The instructor should ask the students to play the module again with the same scenario or if they want to change some parameters that can be done. The students again played the module and completed it.

Once the module is complete the instructor can quickly review the decision-making made by the students along with the results, they achieved across the rounds to compare their progress, trends, and approach towards the module. The emphasis of this plan is to let the students play the module multiple times so they can learn from their previous rounds. The instructors generally ask students to play the module 3 times, whereas the 3rd time is for inculcating concepts.

1. Introduction (5 minutes)
2. Run Simulation (30 to 35 minutes)
3. Debrief (10 minutes)
4. Run Simulation (30 to 35 minutes)
5. Conclusion (8 to 10 minutes)
6. Run Simulation (self-paced at their own time)
7. Short Presentation & Takeaway.

3.1 Concepts that can be Discussed through each decision Areas

I. Observe Phase

Sentimental Analysis

Sentiment analysis is a computational method that interprets and classifies emotions (such as positive, negative, or neutral) within text data, often by analysing social media posts, reviews, customer feedback, and other sources of written content. This technique uses natural language processing (NLP) and machine learning, to evaluate the tone or emotional weight of language, which can provide valuable insights into public opinion, customer satisfaction, and brand perception.

Sentiment Analysis in Design Thinking

Design thinking is a problem-solving framework that prioritizes empathy and a human-centered approach to create innovative solutions. Sentiment analysis is important in design thinking because it provides designers with a deeper understanding of users' emotions, preferences, and pain points by aggregating and analysing their feedback on a large scale.

Here are some key reasons sentiment analysis is important in design thinking:

- a) **Understanding User Needs and Emotions:** Sentiment analysis in the empathy phase helps designers understand users' emotional responses, uncovering needs that may not be immediately visible.
- b) **Enhancing Empathy:** By interpreting the tone of user feedback, designers can empathize with users, gaining a nuanced understanding of unspoken frustrations or desires.
- c) **Prioritizing Pain Points:** Identifying patterns in feedback helps designers focus on key issues and valued features, making it easier to address critical areas first.
- d) **Supporting Ideation:** Real user feedback informs the ideation phase, guiding designers on which features to enhance and where improvements are needed.
- e) **Monitoring and Iterating:** Continuous feedback on prototypes allows designers to adjust solutions based on real-time user sentiment.
- f) **Aligning with Business Goals:** Sentiment analysis helps track how design choices impact brand reputation and customer loyalty, ensuring alignment with business objectives.

Here's a look at what sentiment analysis data might look like, followed by an example of how to interpret it.

Sentiment Analysis Data

Let's consider sentiment analysis data gathered from customer feedback on a fictional fitness tracker. Imagine this feedback data from various channels (e.g., reviews, social media, customer service logs):

Feedback Comment	Sentiment Score	Keywords	Emotion Intensity
"Battery life is really disappointing."	-0.8	disappointing, battery	High
"I love the sleek design, very stylish!"	+0.9	love, stylish	Medium
"It's so frustrating to charge it every day."	-0.7	frustrating, charge	High
"The new tracking features are fantastic!"	+0.8	fantastic, new features	Medium
"Really convenient size and lightweight!"	+0.7	convenient, size, lightweight	Low

"I wish it had better sleep tracking."	-0.3	wish, better, sleep tracking	Low
"The app interface is confusing to use."	-0.6	confusing, interface	High

Let's break down how to interpret this data to guide design decisions.

1. Identify Major Pain Points:

- **Battery Life:** Comments like "really disappointing" and "frustrating to charge every day" appear frequently, with high negative sentiment scores (-0.8, -0.7) and high emotion intensity. The keywords "battery" and "charge" indicate that the battery life is a significant issue, frustrating users and affecting overall satisfaction.
- **Usability Issues:** Feedback with words like "confusing" and "inaccurate" around the app interface and heart rate monitor has moderate negative sentiment (-0.6, -0.5) and medium to high emotional intensity, showing that users struggle with certain functions and accuracy.

2. Identify Positive Sentiment Themes:

- **Aesthetic Design and Size:** Positive comments with words like "stylish," "sleek," and "lightweight" have high sentiment scores (+0.9, +0.7) and moderate intensity, indicating that the product's design and size are appealing to users.
- **Feature Satisfaction:** Sentiment around new tracking features (e.g., "fantastic") scores positively (+0.8), suggesting that users appreciate this addition.

3. Prioritize Design Improvements:

- **Battery Life Enhancement:** Due to high negative sentiment and strong emotion surrounding battery life, the design team should prioritize battery improvements.
- **Simplifying the Interface:** With "confusing" and usability issues scoring negatively and carrying high emotional weight, improving the app interface and making navigation more intuitive could alleviate user frustration.
- **Maintain Positive Aspects:** Positive feedback on the design and new features suggests the team should keep these elements consistent while focusing on refinements.

II. Define Phase

Role of SWOT and Gap Analysis

- **Understanding Strengths and Weaknesses:** This helps teams identify internal advantages and limitations. For example, a tech company might leverage strong R&D capabilities as a strength while acknowledging its limited brand awareness as a weakness.
- **Identifying Opportunities and Threats:** Recognizing external opportunities (such as market trends or new technologies) and threats (like emerging competitors) helps teams anticipate changes and adapt quickly. For instance, when Spotify entered the music streaming market, it analysed the threat posed by piracy and the opportunity to provide a legal, user-friendly alternative.

Gap Analysis:

- **Assessing Current vs. Desired State:** Gap analysis identifies the difference between the existing product/service and customer expectations. For example, a mobile phone company might identify a gap in battery life and address this issue in future product designs.
- **Guiding Iterative Improvement:** By understanding the gaps, teams can prioritize areas for improvement. For instance, a clothing retailer might conduct a gap analysis that reveals a need for better online shopping experiences, leading to enhancements in its e-commerce platform.

III. Ideate Phase

During the Ideation phase, the goal is to generate a large volume of ideas. While not all will become solutions, this stage is about exploring various possibilities and thinking creatively. The emphasis is on quantity over quality, aiming to uncover new perspectives and innovative approaches. It's important to keep the environment judgment-free to encourage free-flowing, unconventional ideas.

Tools/ methods that can be used in the Ideate Phase

Some important tools and methods in the **Ideation Phase** of Design Thinking are as follows:

Tool/Method	Description	Purpose
Brainstorming	A group creativity technique where participants rapidly generate ideas without judgment.	To encourage free-flowing ideas and foster creative thinking
Mind Mapping	A visual diagram to organize and explore ideas by showing relationships between concepts.	To break down complex problems, expand thinking, and visualize connections between ideas.
SCAMPER	A checklist method that encourages thinking of new ways to improve existing ideas through actions like Substitute, Combine, Adapt, Modify, put to another use, Eliminate, and Reverse.	To push thinking beyond conventional boundaries by transforming existing concepts or products.
Sketching/Storyboarding	Creating visual representations of ideas or solutions, often in a step-by-step format.	To help visualize ideas, test different scenarios, and simplify complex concepts.
Role Playing	Participants act out roles or situations to explore how users might interact with a product or service.	To develop empathy and better understand user experiences and needs in context.
Brainwriting	A variation of brainstorming where participants write down their ideas silently before sharing.	To ensure that everyone's ideas are heard and prevent dominant voices from overshadowing others.
Crazy 8s	A fast-sketching exercise where participants create 8 variations of an idea in 8 minutes.	To encourage rapid ideation and push creative thinking by forcing participants to generate diverse ideas quickly.
Reverse Thinking	Encourages considering the opposite of the problem or the solution in order to discover new possibilities.	To challenge assumptions and generate unexpected ideas by thinking from a completely different angle.
Affinity Diagram	Organizes ideas into categories based on common themes or patterns.	To identify key insights, cluster related ideas, and make sense of large amounts of information.

IV. Prototype Phase

Effective Strategies for Managing Prototype Phase Costs

To manage fluctuating costs during the prototype phase while maximizing value, consider the following strategies:

1. Set Clear Objectives and Prioritize Features:

- Focus on the Minimum Viable Product (MVP) by targeting core features that address key user needs, creating a leaner prototype.
- Prioritize high-impact features based on user research to allocate resources effectively.

2. Choose the Right Prototyping Method:

- Start with low-fidelity prototypes (e.g., sketches or wireframes) to gather early insights with minimal cost.
- Use high-fidelity prototypes only for critical features and consider digital simulations to reduce material costs.

3. Utilize Modular Prototyping:

- Break the prototype into testable modules, enabling focused testing and reducing the need for complete redesigns.
- Reuse and iterate on modular components, cutting down on redundancy and costs.

4. Incorporate Rapid Prototyping Technologies:

- Use 3D printing and digital tools (e.g., CAD software) for quick adjustments and reduced material waste.
- Leverage virtual reality (VR) for simulating and testing designs without physical resources.

5. Implement a Flexible Budgeting Approach:

- Set a variable budget with buffer zones for unforeseen expenses and monitor spending in real-time to adjust dynamically.

6. Involve Cross-Functional Teams Early:

- Collaborate with engineers and stakeholders early to identify cost drivers and feasibility issues.
- Conduct early user feedback sessions to validate assumptions and avoid costly rework.

7. Leverage Partnerships and Outsourcing:

- Build relationships with suppliers to gain access to cost-effective materials and discounts.
- Outsource specialized components to reduce in-house production costs.

8. Adopt Agile and Lean Methodologies:

- Use agile development cycles with incremental improvements to manage resources efficiently.
- Apply lean principles to eliminate waste and streamline the prototyping process.

9. Perform Regular Cost-Benefit Analysis:

- Assess the ROI of each prototype iteration to ensure improvements align with project goals.
- Focus on learning value from each iteration, reducing risk, and validating assumptions rather than seeking perfection.

V. Execute Phase

The **Execute Phase** demands careful consideration of how the product will be presented, priced, promoted, and sold. When deciding on elements like **price, promotion, warranty, and channel investment**, ensure they align with the product's value proposition, market positioning, and customer expectations. Striking the right balance in these decisions is crucial for launching a product that not only meets customer needs but also succeeds in a competitive market. Each decision must reflect a deep understanding of the target audience, competitive landscape, and overall business goals to drive long-term success.

The key thoughts and considerations to keep in mind when making these decisions in real case scenarios are as follows:

Key Decision	Considerations
Price	• Customer Value Perception: The price should reflect the product's value based on features, quality, and uniqueness. • Competitive Landscape: Analyse competitor pricing and adjust accordingly, especially if your product has unique features. • Cost and Margin Considerations: Ensure the price covers production, distribution, and marketing costs with healthy profit margins. • Pricing Strategy: Decide between penetration pricing (low price to gain market share) or skimming pricing (high price for higher margins).
Advertising Budget Allocation	• Target Audience: Tailor advertising strategies to demographic preferences • Advertising Channels: Choose the best channels based on target audience behaviour. • Budget Allocation: Allocate the budget efficiently to maximize ROI, considering each channel's cost and potential return.
Warranty Period	• Product Reliability: Longer warranties for high-quality, reliable products; shorter ones for lower-cost or lower-risk products. • Customer Expectations: Offer competitive warranties in line with industry standards to build trust. • Cost Implications: Longer warranties may incur higher costs for repairs or replacements. • Brand Positioning: A strong warranty (e.g., lifetime) can position the product as premium; a shorter one may work for value-oriented products.
Investment in Channels	• Channel Fit: Align sales channels with the target audience's purchasing preferences • Market Reach: Consider each channel's ability to expand market reach • Customer Experience: Choose channels that offer the best customer experience • Cost and ROI: Consider cost of investment in each channel and expected returns. Channel Synergy: Some channels, like combining e-commerce with telecom partnerships, may work better together for enhanced reach.

3.2 Case Customization

The case can be customized based on the faculty's requirements, providing the flexibility to enable or disable specific decision tabs within each module. Additionally, various parameters, such as market, financial statements, etc., can be adjusted to meet specific needs.

3.3 Suggested Discussion Starters

1. Why is user research so important in the design thinking framework, and what methods work best for gathering insights?
2. How does the empathy phase of design thinking influence the product development process?
3. How can sentiment analysis enhance the empathy phase in design thinking?
4. In what ways can design thinking help companies better align their products with customer needs?
5. How can the concept of “fail fast, fail cheap” be implemented in the prototyping phase to avoid major setbacks?
6. What is the role of ideation in design thinking, and how do you ensure that the ideas generated are viable and feasible?
7. Why is iteration a key aspect of design thinking, and how can it be managed efficiently in real-time?
8. How can teams avoid confirmation bias when interpreting user feedback in the design thinking process?
9. How do you ensure that a solution created through design thinking is scalable and adaptable over time?
10. How can design thinking be used to create products that resonate emotionally with users, not just functionally?
11. What are some common challenges teams face when transitioning from the prototyping phase to the execution phase in design thinking?
12. How can you ensure that a prototype reflects the actual needs and desires of your target audience?
13. How do you measure the success of a design thinking process or solution in the real world?
14. How can design thinking help companies pivot their product offerings based on changing market conditions or user feedback?
15. How important is it for companies to involve customers throughout the design thinking process, and how can they do so effectively?